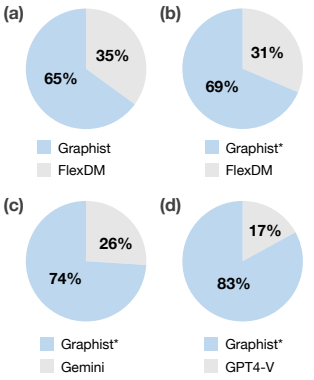
Table 2: GPT-4V Eval on Crello. The table demonstrates the performance of different methods on the Crello dataset. Graphist is Graphist-Base built upon Qwen1.5-7B. The scores on the left are GPT-4V rating and IOPR_{avg}, while the chart on the right showcases a comparative evaluation using GPT-4V voting against the FlexDM [36], Gemini-1.5-Pro (Gemini) and GPT-4V. Flex-DM is compared based on the GLG task (a, b), while Gemini-1.5-Pro and GPT-4V are based on the HLG task (c, d).

Method	Task	S_{DL}	S_{GI}	S_{IO}	S_{TV}	IOPR
Flex-DM	*GLG*	5.43	6.13	4.69	4.60	-
GPT-4V	*GLG*	5.10	5.83	4.54	3.84	-
Gemini	*GLG*	4.87	5.62	4.39	4.09	-
Graphist	*GLG*	5.60	6.64	4.84	4.86	-
Graphist∗	*GLG*	**5.72**	**6.70**	**5.03**	**5.37**	-
GPT-4V	*GLG*	4.19	4.66	3.71	3.25	0.45
Gemini	*GLG*	5.06	5.94	4.33	4.21	0.70
Graphist	*HLG*	5.66	6.60	5.02	4.93	0.96
Graphist∗	*HLG*	**5.85**	**6.90**	**5.10**	**5.24**	**0.97**



5.3 Comparison with SoTA

We trained our model using the approach outlined in Section 4.2 and compared it to other state-of-the-art methods. In the following comparison, we represent the model trained only on public datasets as "Graphist", while "Graphist*" denotes the model trained on in-house data in addition to public datasets.

Results in GLG & HLG tasks. The Crello dataset evaluation underscores the dominance of Graphist* and Graphist over Flex-DM across both tasks. In the GLG task, Graphist* achieved the highest scores in three out of four domains, with marked gains in S_{GI}, attaining a commanding 6.70, and similarly impressive performance in S_{DL} (5.72) and S_{IO} (5.03). In S_{TV}, Graphist* rendered a commendable score, highlighting the method's robustness in maintaining text readability amidst complex designs. Shifting to the more challenging HLG task, Graphist* not only continued its strong showing but exhibited even more dramatic enhancements, scoring uniformly higher over the counterpart, Flex-DM. Noteworthy is the peak score of 5.85 for S_{DL}, underscoring Graphist*'s proficiency in clean and balanced layout generation. This version also excelled notably in graphic-image synergy with a top score of 6.90 in S_{GI} and innovation with a superior score of 5.10 in S_{IO}. Moreover, the S_{TV} metric remains above Flex-DM, reflecting the model's diligence in preserving text clarity across design variations. We also evaluate the performance of top-tier proprietary linear mixed models, specifically Gemini-1.5-Pro (Gemini) and GPT-4V, on both tasks. Although they offer greater versatility, there is a significant performance gap compared to our Graphist on layout-centric tasks, such as GLG and HLG.

Fig. 4: Results visualization of the GLG task on the Crello dataset. The results for Flex-DM were derived from their open-source code, whereas the results for GPT-4V and Gemini-1.5-Pro are obtained in zero-shot manner.

Regarding the evaluation details, there may be situations where GPT-4V and Gemini-1.5-Pro cannot correctly infer the required JSON format. We have prepared format post-processing for their common return result types, and we give these two methods five inference opportunities for each test case. If all are unsuccessful, then skip. In the end, GPT-4V retained 120 results, and Gemini-1.5-Pro retained 240 results. We reported their scores on these subsets.

In the visualization of the GLG outcomes in Figure 4, Graphist outperforms competitors including Flex-DM, GPT-4V, and Gemini-1.5-Pro. The other methodologies grapple with challenges such as text overlap and image distortion. Furthermore, we evaluated real-world design elements, contrasting the performance of Gemini-1.5-Pro with Graphist, as depicted in Figure 5. For complex design tasks involving over ten layers, our approach yield more refined design outcomes. Although Gemini-1.5-Pro managed to accomplish the HLG task, the design acuity is markedly distinct from that of Graphist.

Results on CGL-V2. Besides the HLG and GLG tasks, our model can adapt to various other design tasks through flexible configurations. We conducted Content-Aware and Content-Agnostic layout generation experiments on the CGL-

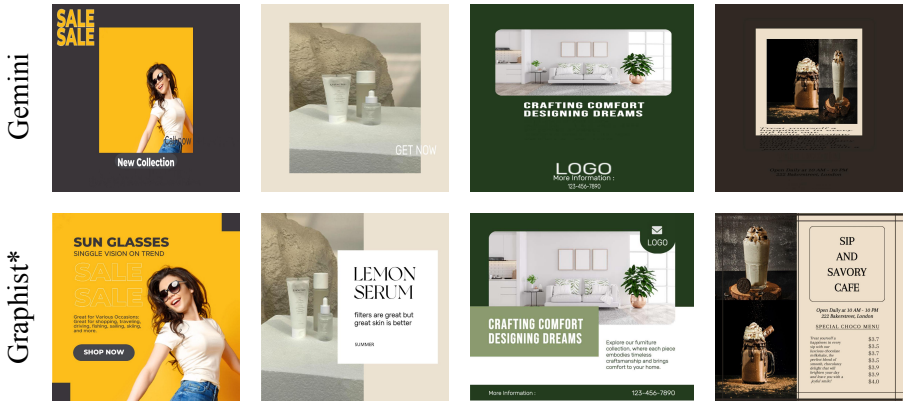


Fig. 5: Comparison with SoTA method on the HLG task with real-world design elements. Both Gemini-1.5-Pro and Graphist models are tasked with HLG using identical real-world design elements. The outcomes indicate superior design quality achieved by our Graphist* when compared with Gemini-1.5-Pro.

Table 3: Comparison with content-aware methods on CGL-V2. In this table, we compare Graphist based on Qwen1.5-7B [3] against other content-aware methods. Content-aware refers to generating a layout based on the input background image and text content. C-R means Composition-relevant.

Method	C-R measures		Graphic measures		
	R_{com} ↓	R_{occ} ↑	R_{ali} ↓	R_{ove} ↓	R_{und} ↑
ContentGAN [47]	31.930	**1.000**	0.009	0.065	0.840
CGL-GAN [101]	16.040	0.875	0.007	0.081	0.732
RADM [46]	**10.260**	0.997	0.008	0.046	0.983
Graphist*	12.150	**1.000**	**0.004**	**0.002**	**0.996**

V2 dataset based on Qwen1.5-7B and following the experiment settings and evaluation metrics of [101].

In the Content-Aware layout design, from Table 3, Graphist stands out with the lowest R_{com} of 12.150, the highest R_{occ} tied with ContentGAN at 1.000, and leading performances in all graphic measures ($R_{ali} = 0.004, R_{ove} = 0.002, R_{und} = 0.996$), indicating superior grapic visual balance, non-emptiness, alignment, overlap avoidance, and underlay success.

Similarly, within Content-Agnostic layout task, Graphist again showcases its robust capability by achieving the best R_{com} score of 7.074 and perfect R_{occ} score, which is illustrated in Table 4. It also surpasses competitors in graphic measures, underpinning its strength in alignment ($R_{ali} = 0.0003$), minimizing elements overlap ($R_{ove} = 0.001$), and maximizing the informative underlay usage ($R_{und} = 1.0$). These results cement Graphist as a versatile and potent solution